

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns

df = pd.read_csv('Customer Churn.csv')
df.head()
```

```
{"type": "dataframe", "variable_name": "df"}
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                7043 non-null   object
2   SeniorCitizen         7043 non-null   int64
3   Partner               7043 non-null   object
4   Dependents            7043 non-null   object
5   tenure                7043 non-null   int64
6   PhoneService          7043 non-null   object
7   MultipleLines         7043 non-null   object
8   InternetService       7043 non-null   object
9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
13  StreamingTV           7043 non-null   object
14  StreamingMovies       7043 non-null   object
15  Contract              7043 non-null   object
16  PaperlessBilling      7043 non-null   object
17  PaymentMethod         7043 non-null   object
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   object
20  Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
#replacing blanks with 0 as tenure is 0 and no total charges recoded
```

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")
df["TotalCharges"] = df["TotalCharges"].astype("float")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
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17	PaymentMethod	7043 non-null	object
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	float64
20	Churn	7043 non-null	object

```
dtypes: float64(2), int64(2), object(17)
```

```
memory usage: 1.1+ MB
```

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.describe()
```

```
{"summary":{"name":"df",\n"rows": 8,\n"fields": [\n{\n"column": "SeniorCitizen",\n"properties": {\n"dtype": "number",\n"std": 2489.9992387084,\n"min": 0.0,\n"max": 7043.0,\n"num_unique_values": 5,\n"samples": [\n0.1621468124378816,\n1.0,\n0.36861160561002687\n],\n"semantic_type": "\",\n"description": "\"\n}],\n{\n"column": "tenure",\n"properties": {\n"dtype": "number",\n"std": 2478.9752758409018,\n"min": 0.0,\n"max": 7043.0,\n"num_unique_values": 8,\n"samples": [\n32.37114865824223,\n29.0,\n7043.0\n],\n"semantic_type": "\",\n"description": "\"\n}]}
```

```

n    },\n    {\n        \"column\": \"MonthlyCharges\", \n        \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 2468.7047672837775, \n            \"min\": 18.25, \n            \"max\": 7043.0, \n            \"num_unique_values\": 8, \n            \"samples\": [\n                64.76169246059918, \n                70.35, \n                7043.0\n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    },\n    {\n        \"column\": \"TotalCharges\", \n        \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 3122.5732655623974, \n            \"min\": 0.0, \n            \"max\": 8684.8, \n            \"num_unique_values\": 8, \n            \"samples\": [\n                2279.7343035638223, \n                1394.55, \n                7043.0\n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n        } \n    } \n    ],\n    \"type\": \"dataframe\"}

```

```
df[\"customerID\"].duplicated().sum()
```

```
np.int64(0)
```

```

def conv(value):
    if value == 1:
        return \"yes\"
    else:
        return \"no\"

```

```
df['SeniorCitizen'] = df[\"SeniorCitizen\"].apply(conv)
```

#convert 0 and 1 values of senior citizen to yes or no to make it easier to understand

```
df.head(10)
```

```
{\"type\": \"dataframe\", \"variable_name\": \"df\"}
```

```

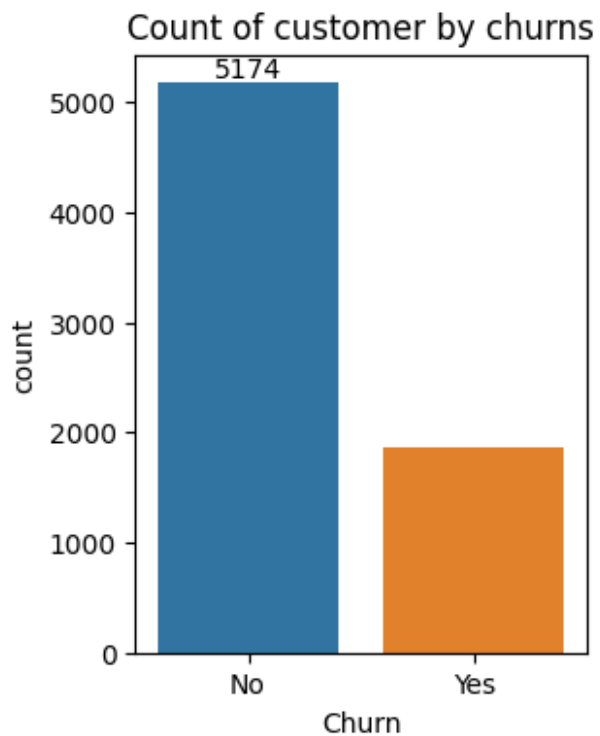
plt.figure(figsize=(3,4))
ax = sns.countplot(x = 'Churn', data = df , hue= \"Churn\")

```

```

ax.bar_label(ax.containers[0])
plt.title(\"Count of customer by churns\")
plt.show()

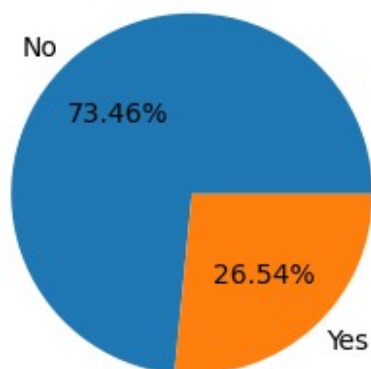
```



```
plt.figure(figsize=[3,4])
gb = df.groupby('Churn').agg({'Churn':"count"})

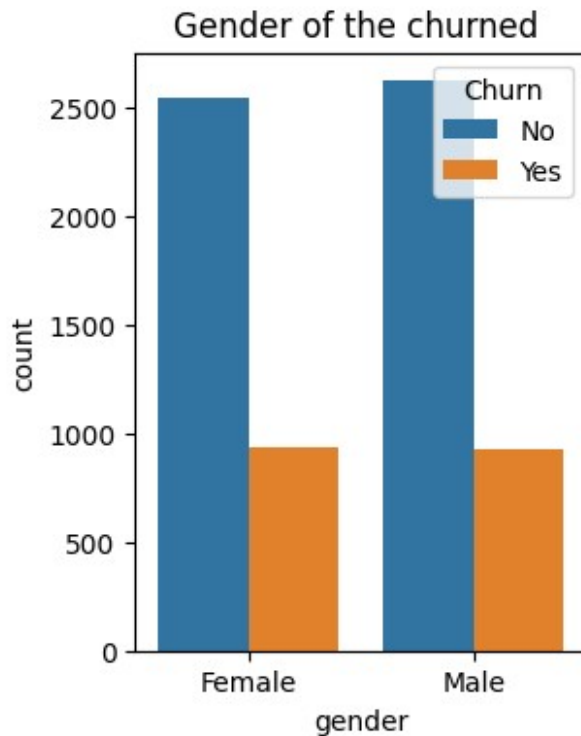
plt.pie(gb['Churn'] , labels = gb.index, autopct = "%1.2f%%")
plt.title("Percentage of churned customers", fontsize = 12)
plt.show()
```

Percentage of churned customers

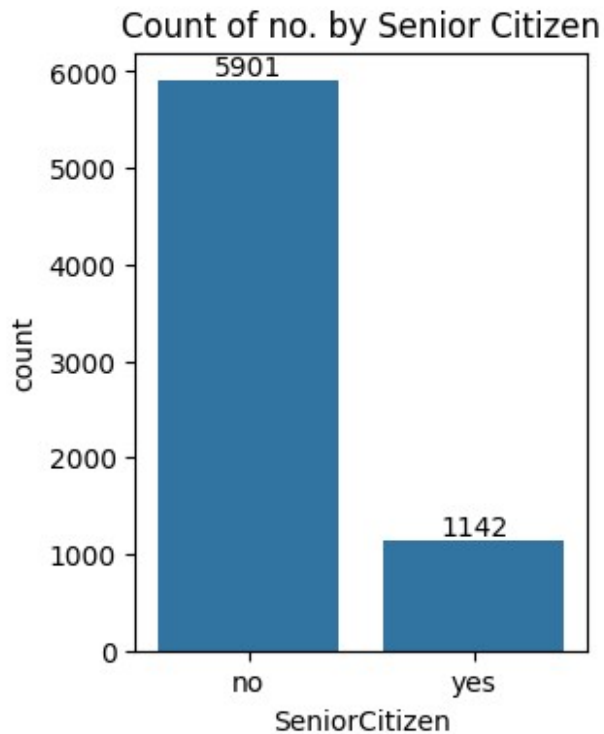


from thus given pie char we can conclude thta
26.54 customers are churned out

```
plt.figure(figsize=(3,4))  
sns.countplot(x= "gender", data=df , hue= "Churn")  
plt.title("Gender of the churned ")  
plt.show()
```



```
plt.figure(figsize=(3,4))  
ax = sns.countplot(x= "SeniorCitizen", data = df )  
ax.bar_label(ax.containers[0])  
plt.title("Count of no. by Senior Citizen")  
plt.show()
```

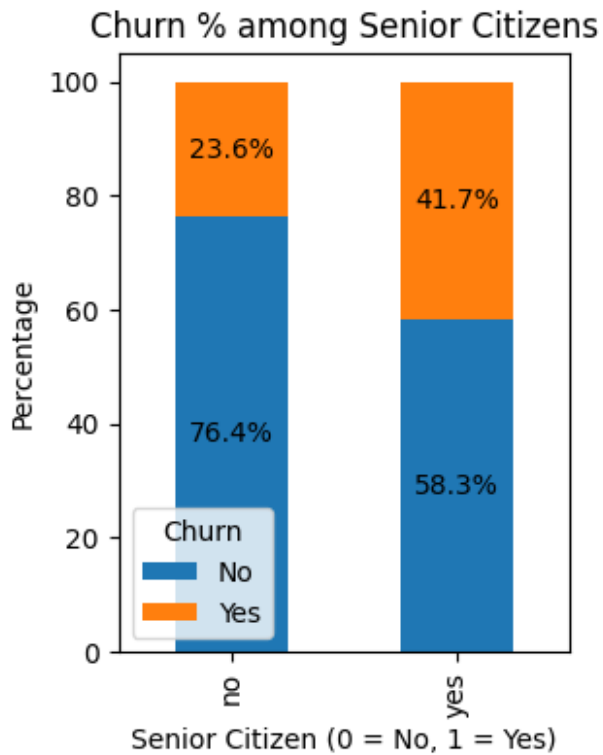


```
ct = pd.crosstab(df['SeniorCitizen'], df['Churn'], normalize='index')
* 100

ax = ct.plot(kind='bar', stacked=True, figsize=(3,4))

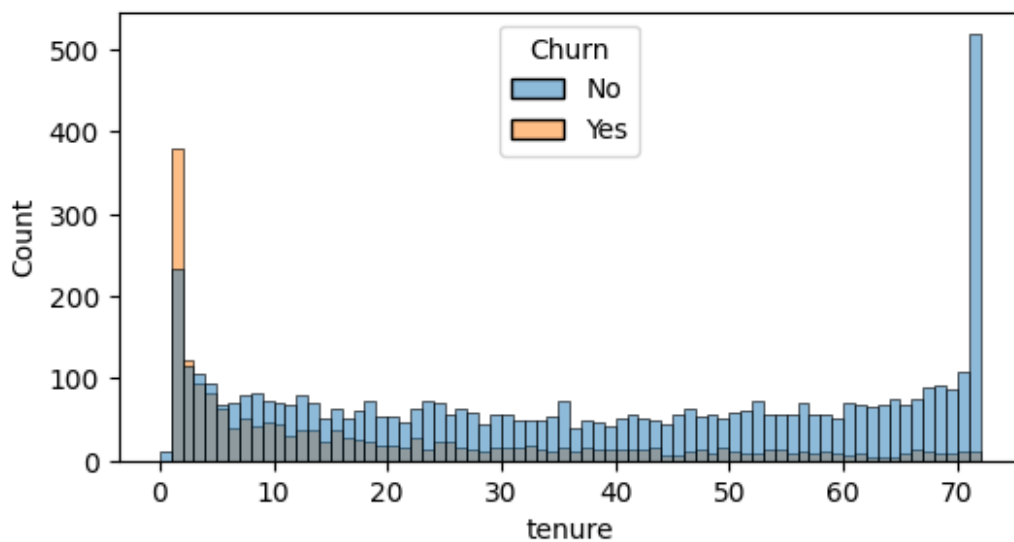
for i in range(len(ct)):
    bottom = 0
    for col in ct.columns:
        value = ct.loc[ct.index[i], col]
        ax.text(i, bottom + value/2, f'{value:.1f}%',
                ha='center', va='center')
        bottom += value

plt.title("Churn % among Senior Citizens")
plt.xlabel("Senior Citizen (0 = No, 1 = Yes)")
plt.ylabel("Percentage")
plt.legend(title="Churn")
plt.show()
```



Comapritve a greated % of people in senior citizen category have churned

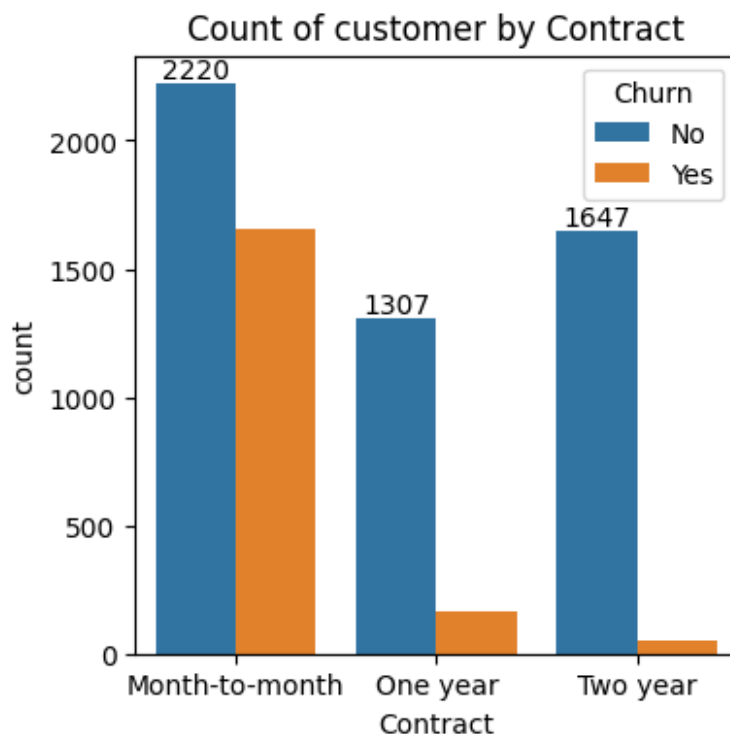
```
plt.figure(figsize=[6,3])
sns.histplot(x = "tenure", data = df , bins = 72 , hue = "Churn")
plt.show()
```



#People who have used our services for a long time have stayed and people who have used our services

1 or 2 months have churned

```
plt.figure(figsize=(4,4))
ax = sns.countplot(x= "Contract", data = df , hue= "Churn")
ax.bar_label(ax.containers[0])
plt.title("Count of customer by Contract")
plt.show()
```



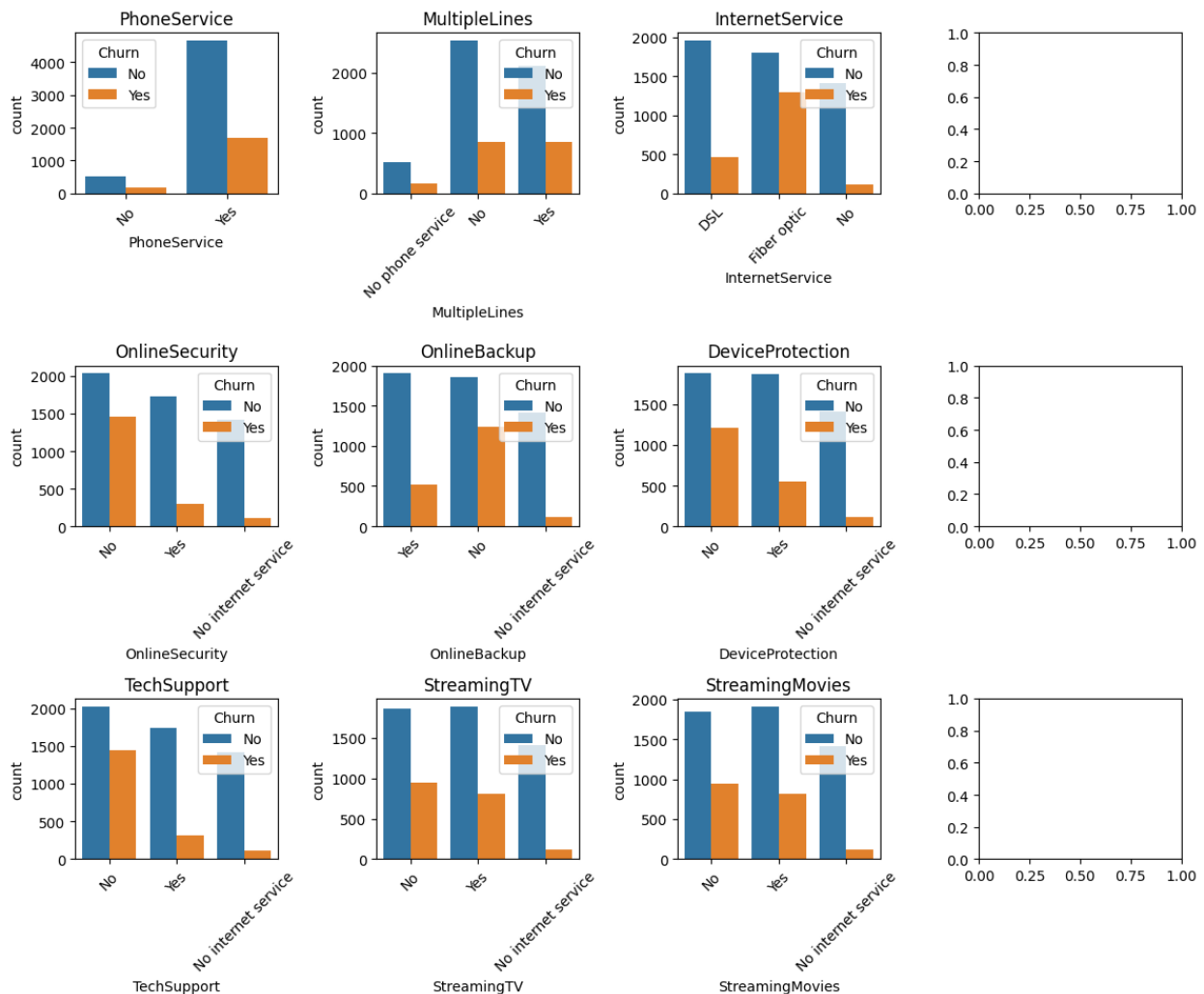
```
df.columns.values
array(['customerID', 'gender', 'SeniorCitizen', 'Partner',
      'Dependents',
      'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
      'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
      'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
      'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
      'TotalCharges', 'Churn'], dtype=object)

cols = ['PhoneService', 'MultipleLines', 'InternetService',
        'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
        'TechSupport', 'StreamingTV', 'StreamingMovies']
```

```
fig, axes = plt.subplots(3,4 , figsize=(12,10))

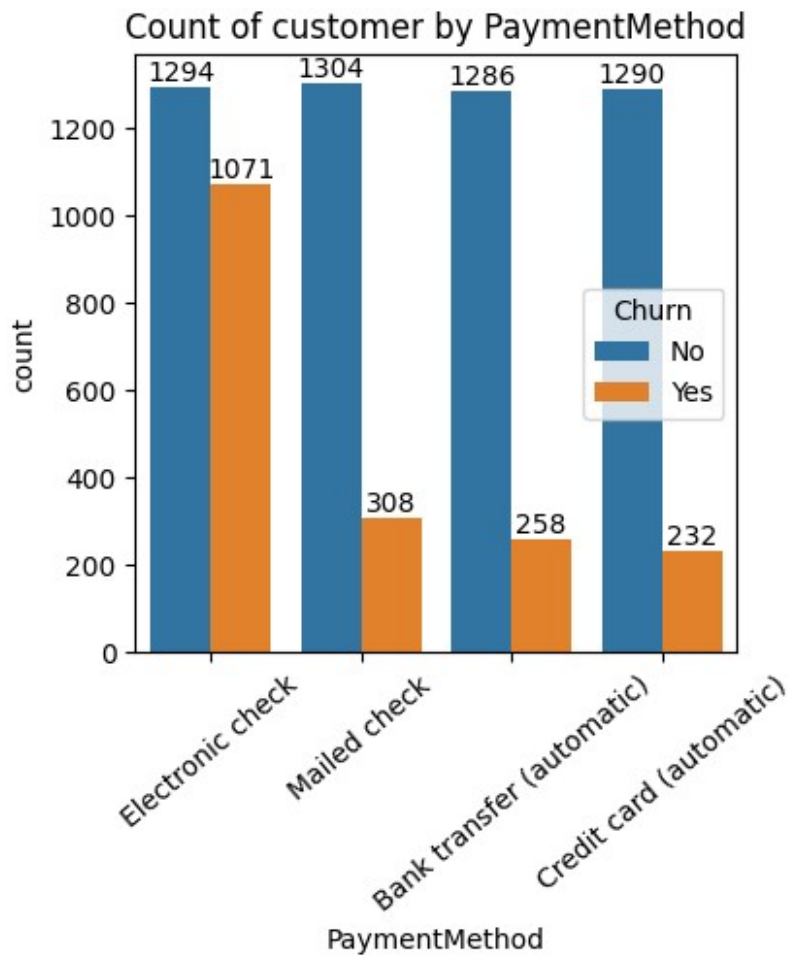
for i, col in enumerate(cols):
    ax = axes[i//3, i%3]
    sns.countplot(x=col, data=df, ax=ax , hue = df["Churn"])
    ax.set_title(col)
    ax.tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.show()
```



#Online Backup and Device Protection are key drivers of customer loyalty, showing significantly lower churn rates than average. In contrast, basic communication and entertainment services, such as Phone Service and Streaming TV, do not notably improve retention and are often linked to more price-sensitive plans. Ultimately, customers subscribing to security-focused bundles are the most stable, while those with only entertainment or phone services are more likely to leave

```
plt.figure(figsize=(4,4))
ax = sns.countplot(x= "PaymentMethod", data = df , hue= "Churn")
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("Count of customer by PaymentMethod")
plt.xticks( rotation = 40)
plt.show()
```



#customer is likely to churn when he is using electronic check as a payment method